# 3008ICT Deep Learning

# Lab 10 (Pre-training, BERT, and Tokenisation)

# Questions and Discussions

**Part 1: True / False Questions**

1. The "Bitter Lesson" in AI suggests that complex, hand-engineered approaches will always outperform simple algorithms in the long run.

True / False?

2. When evaluating a language model, a lower perplexity score indicates better performance.

True / False?

3. BERT is fundamentally an autoregressive model capable of naturally generating text left-to-right.

True / False?

4. Modern pre-trained language models rely heavily on subword tokenisation instead of standard word-level embeddings like GloVe.

True / False?

Part 2: Multiple Choice Questions

5. Which of the following best describes how BERT achieves bidirectionality?

a. It concatenates two independent unidirectional LSTMs (forward and backward).
b. It uses a standard autoregressive objective but reads the text from right-to-left.
c. It utilises the Transformer architecture and a Masked Language Modeling (MLM) objective.
d. It uses Byte Pair Encoding to reverse the order of characters.

6. What is the primary purpose of the [CLS] token in BERT?

a. To act as a boundary separating two different sentences.
b. To predict the masked words in the sequence.
c. To represent the aggregated sequence-level information for classification tasks.
d. To indicate the end of a generated sequence.

7. How does BERT handle extractive Question Answering tasks (like SQuAD)?

a. It generates the answer sequence left-to-right.
b. It predicts the start and end indices of the answer span within the provided passage.

c. It classifies the entire passage as either 'True' or 'False'.
d. It retrieves the answer from an external knowledge base.

8. Why did modern NLP models move away from strictly character-level models?

a. Character-level models cannot handle out-of-vocabulary terms.
b. They resulted in sequences that were too long, making computation extremely expensive.
c. They required massive softmax matrices that slowed down training.
d. They overfit easily on small datasets.

Part 3: Short & Long Answer Questions

9. Explain the mechanism of Byte Pair Encoding (BPE). How does it construct its vocabulary?

10. Contrast the bidirectionality of ELMO with the bidirectionality of BERT. Why is BERT's approach considered "deeply bidirectional"?

Answers:

**Part 1: True / False**

1. False - The "Bitter Lesson" states that simple algorithms leveraging scaling (more data/compute) outperform hand-engineered approaches.

2. True - Perplexity is the exponential of the average negative log-likelihood; lower values mean the model assigns higher probability to the true data.

3. False - BERT is designed for bidirectional "analysis" tasks (like classification or extraction) and cannot natively generate text left-to-right in an obvious/efficient way.

4. True - Models usually use between 50k and 250k subword pieces rather than classical word embeddings.

**Part 2: Multiple Choice**

5. c - It utilises the Transformer architecture and a Masked Language Modeling (MLM) objective to prevent "cheating" while looking at both directions.

6. c - To represent the aggregated sequence-level information for classification tasks.

7. b - It predicts the start and end indices of the answer span within the provided passage.

8. b - They resulted in sequences that were too long, making computation extremely expensive (especially for self-attention mechanisms).

**Part 3: Short & Long Answer**

9. BPE Mechanism:

BPE starts by treating every individual character/byte as its own symbol. It counts the co-occurrences of adjacent symbols in the training corpus and iteratively merges the most frequent pair into a single new subword token. It repeats this process for a specified number of merges (e.g., 8000), effectively creating a vocabulary of frequent whole words and common word parts.

10. ELMO vs BERT Bidirectionality:

ELMO approximates bidirectionality by training a forward-facing language model and a backward-facing language model entirely independently, then concatenating their hidden states. BERT is "deeply bidirectional" because it uses a single Transformer where self-attention allows every token to attend to every other token (left and right) simultaneously across all layers. To prevent the model from “cheating" (seeing the word it's trying to predict), BERT masks out 15% of the input tokens.